Table 11.1 provides an overview of possible combinations and transitions from one signal type to another, with traces representing the cross-services execution, metrics representing the numerical aggregates (e.g., percentage of errors), and logs as the signal type representing a single service along the request path. Note that these transitions are not specific to a format, such as OpenTelemetry.

Table 11.1 Overview of signal correlations

From/To	Traces	Metrics	Logs
Traces	Related spans	Trace to representative metrics	Trace to log lines from services
Metrics	Metrics to representative spans	Related metrics	Metrics to logs from services
Logs	Logs to spans from services	Logs to relevant metrics	Related logs

The term *correlation path* in the following subsections refers to the transitioning from one signal type to another, correlated one. This may include more than one transition. Let's dissect table 11.1 now, discussing specific correlations by signal type.

11.1.2 Correlating traces

In this first case, the starting points of correlation paths are traces, representing cross-service execution, covering synchronous services along the request path and, potentially, async services, such as message brokers. From here, we will transition to specific correlations.

To traces

Traces–traces correlation refers to finding related spans by environment or service. For example, you can use the Span Links feature (http://mng.bz/0KNW) in OpenTelemetry, which allows you to associate a span with one or more other spans, defining a causal relationship for asynchronous fire-and-forget workloads, where a parent span doesn't wait for its children to close before it closes itself. This is particularly useful for async services such as message buses and message queues, including Apache Kafka (https://kafka.apache.org/) and Amazon EventBridge (https://aws.amazon.com/eventbridge/).

To metrics

Traces–metrics correlation means the input is a trace representing cross-service execution and the output consists of relevant metrics from services contributing to the spans of the input trace.

To logs

Traces–logs correlation means the input is a trace representing cross-service execution and the output consists of log lines from services contributing to the spans of the